Pradnya Sushil Shinde

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EDUCATION

Worcester Polytechnic Institute **Aug 2023 - May 2025**

*Master of Science in Robotics Engineering (**GPA: 4.0/4.0**)* *Worcester, MA*

Relevant Coursework: Computer Vision, Vision-based Robotic Manipulation, Motion Planning, Advanced Robot Navigation

Symbiosis Skills and Professional University **Aug 2019 - May 2023**

*Bachelor of Technology in Mechatronics Engineering (**GPA: 9.128/10.0**)* *Pune, India*

SKILLS

Languages: Python, C++, C

Libraries, Tools, and Software: Linux OS, ROS, ROS2, Nav2, Gazebo, Docker, RViz, OpenCV, PCL, SLAM Toolbox, NumPy, Matplotlib, Sklearn, PyTorch, Eigen, CUDA, Git, CMake, Blender, Arduino, MATLAB, Solidworks.

Hardware: Arduino, ESP32, ESP8266, Raspberry Pi, LiDAR

EXPERIENCE

Software Engineer I, Robotics **Jun '25 - Present**

Nightingale Intelligent Systems *Newark, CA*

- Contributing to and co-owning a ROS-based software autonomy stack for aerial vehicles; provisioned and integrated software, network, and system capabilities across 20+ base station and drone systems to ensure reliable distributed communication and deployment readiness.
- Supported software integration and BVLOS waiver evidence collection for aerial vehicles, while diagnosing and resolving live autonomy software issues through direct customer support to improve system reliability.

Robotics Software Engineer Intern **Jan '25 - Apr '25**

Nightingale Intelligent Systems *Newark, CA*

- Optimized UAV path planning by implementing and tuning a Multi-Heuristic A* algorithm, improving route efficiency and planner robustness.
- Engineered a Dockerized ROS–Gazebo simulation framework, streamlining codebase integration and dependencies.

Graduate Student Researcher **Aug '24 - May '25**

Worcester Polytechnic Institute *Remote*

- Analyzed distributed **SLAM** techniques for agricultural robotics using Kimera, an open-source library for real-time metric-semantic localization and mapping.
- Designed and built a data acquisition pipeline to evaluate the library's applicability and performance in agricultural environments.

Robotics Software Engineer Intern **Aug '24 - Dec '24**

Celltrio Inc. *Fremont, CA*

- Performed kinematic and motion analysis on a 7-axis SCARA robot to enable accurate tool transformations and end-effector control.
- Reviewed robot codebase to improve barcode scanning efficiency and motion performance, implementing code enhancements that increased operational efficiency.

Robotics Software Engineer Intern **May '24 - Aug '24**

National Robotics Engineering Center *Pittsburgh, PA*

- Deployed a LiDAR-based localization and mapping pipeline in ROS 2, enhancing warehouse mobile manipulator navigation.
- Conducted unit testing and hardware-in-the-loop (HIL) validation, ensuring reliable sensor integration and system performance.

Robotics Engineer Intern **Jun '22 - May '23**

Technodune Pvt. Ltd. *Pune, IN*

- Led software testing and embedded systems integration for the "**Wirelessly Controlled Bionic Arm**" and "**Real-Time Robot Localization using Vision**" projects using **Python** and **C/C++**.
- Contributed to developing: A 7DOF robotic manipulator with wireless gesture control and web interface; A real-time robot localization platform with Fiducial markers and Yolo V5, achieving **85**% accuracy in robot pose detection.

PROJECTS

Einstein Vision: A Perception Pipeline for Autonomous Vehicles GitHub, Python

- Integrated lane detection, pedestrian pose estimation, road sign recognition, depth estimation, object detection, and scene classification into a unified perception stack, enabling seamless environmental visualization for an autonomous vehicle.

Motion and Trajectory Tracking of Aerial Vehicle GitHub, Python

- Implemented state estimation and global positioning algorithms such as **EKF**, **UKF**, **Particle Filter** and **GNSS/INS** to track the motion of an aerial vehicle effectively.

SfM: Structure from Motion GitHub, Python

- Engineered a robust a Structure from Motion (SfM) system to reconstruct 3D scenes and estimate camera poses from a monocular camera using images from multiple viewpoints.

3D Scene Reconstruction using NeRF GitHub, Python

- Developed a **NeRF**-based 3D scene reconstruction pipeline using a 5D coordinate system and **MLP** to optimize scene rendering from 2D images, and applied differentiable rendering to minimize residuals between synthesized and ground truth images.

Zhang's Camera Calibration: Optimizing Distortion Correction GitHub, Python

- Computed camera Intrinsic and Extrinsic Parameters by minimizing image distortion and reprojection error to perform Zhang's **Camera Calibration** method on checkerboard images.

Homography Estimation:Classical and Deep Leaning Methods GitHub, Python

- Designed a panorama stitching algorithm with corner detection, ANMS, feature matching, and **RANSAC**; trained CNN with TensorDLT and STN to predict homography, achieving $\mathbf{3.5-5.0}\%$ RMSE.

Probability-based Edge Detection: A Classical Approach to Boundary Detection GitHub, Python

- Developed a probability-based edge detection algorithm using filter banks, half-disc masks, k-means clustering for texture, brightness, and color features, and integrated with Sobel and Canny baselines for enhanced boundary detection.

Language Embodied Indoor Navigation GitHub, Python

- Developed a language-guided indoor navigation pipeline that converted spoken commands into navigation intents and executed goal-directed motion using **A*** and **RRT*** planners in Habitat, with RGB, depth, and semantic sensing for environment awareness.

Rigid Body Motion Planning GitHub, Python

- Developed and optimized **RRT**, **RRT***, **Informed RRT***, and **Probabilistic Roadmap**(PRM) based path planners tailored to a point robot, a 2D omnidirectional robot, and a 4-link kinematic chain.

Vision-based Object Grasping for Robotic Manipulation assuming Symmetry GitHub, C++

- Engineered a vision-based grasp pose analysis algorithm for robotic manipulation by processing depth point clouds, estimating symmetry for grasp point computation, using **Gazebo** and **ROS 2** for simulation.

PUBLICATION

Paper Title: "Marathi Sign Language Hand Gesture Recognition Using Accelerometer and 3D Printed Gloves" Paper
Status: Published in 2022 14th International Conference on Computational Intelligence and Communication Networks (CICN)
Publication Date: January 13, 2023